UUM
Universiti Utara Malaysia

FINAL EXAMINATION
FIRST SEMESTER 2023/2024 SESSION

COURSE CODE/NAME	:	STID3113/RESEARCH METHOD IN IT
DATE	:	19 FEBRUARY 2024 (MONDAY)
TIME	:	9.00 A.M. – 11.30 A.M. (2 1/2 HOURS)
VENUE	:	DPC 3 & DPC 4 STML

INSTRUCTIONS :

1. This examination paper has **FIVE (5)** case studies in **FOURTEEN (14)** printed pages excluding the cover page.
2. Answer **ALL** questions in the space provided.
3. You are **NOT ALLOWED** to remove the examination paper from the examination hall.

MATRIC NO.: _____ (in words) | | | | | | | (in numbers)

IDENTIFICATION CARD NO.: | | | | | | | | | | | | |

LECTURER NAME: _____

GROUP: | | **TABLE NO.:** | | | | |

DO NOT OPEN THIS EXAMINATION PAPER UNTIL INSTRUCTED

CASE STUDIES (100 MARKS)

INSTRUCTION: Answer **ALL** questions in the space provided.

CASE 1

Given the following research problem:

We are living in a world which is continually evolving and where modern conflicts have moved to the cyber domain. In its 2010 Strategic Concept, NATO affirmed its engagement to reinforce the defence and deterrence of its state members. In this light, it has been suggested that the gamification of training and education for cyber security will be beneficial. Although serious games have demonstrated pedagogic effectiveness in this field, they have only been used in a limited number of contexts, revealing some limitations. Thus, it is argued that serious games could be used in informal contexts while achieving similar pedagogic results. It is also argued that the use of such a serious game could potentially reach a larger audience than existing serious games, while complying with national cyber strategies. To this end, a framework for designing serious games which are aimed at raising an awareness of cyber security to those with little or no knowledge of the subject is presented. The framework, based upon existing frameworks and methodologies, is also accompanied with a set of cyber security skills, itself based upon content extracted from government sponsored awareness campaigns, and a method of integrating these skills into the framework. Finally, future research will be conducted to refine the framework and to improve the set of cyber security related skills in order to suit a larger range of players. A proof of concept will also be designed in order to collect empirical data and to validate the effectiveness of the framework.

Based on the research problem described in the case study, answer the following questions:

1. Identify the problem statement.

(2 marks)

2. State **ONE (1)** appropriate research question.

(2 marks)

3. Based on the research question identified in Question 2, should the research be categorized as basic or applied? Provide justification supporting the research classification.

(2 marks)

4. Describe **TWO (2)** important subtopics to be covered when performing your Literature Review.

(4 marks)

CASE 2

Privacy concerns represent a significant obstacle to the widespread adoption of online technology adoption. However, when consumers are facing personal risks (being ill) and environmental risks (pandemic), the effect of privacy concerns on continued use intention of telemedicine is unknown. The large user pool of virtual visits during COVID-19 provides a great opportunity to understand consumers' privacy concerns when facing personal and environmental risks. This research investigates how patients weigh personal risks (e.g., illness) and environmental risks (e.g., pandemic) against privacy concerns from the lens of Privacy Calculus Model and Protection Motivation Theory. An online survey was conducted to collect data. Analysis indicates that first-time telepatients (82% of respondents) have greater privacy concerns than repeat users. Findings also indicate that patients who are female and have some college education or less reported greater privacy concerns. Interestingly, privacy concerns are positively related to continued use intention. This result holds when satisfaction and user characteristics are controlled. When consumers are ill, privacy concerns still play an important role in telemedicine adoption. However, under environmental risks such as the COVID-19 pandemic, privacy concerns do not negatively impact their continued use intention, and satisfaction is positively associated with continued use intention.

Assuming you are given a research grant to conduct research related to the case presented, prepare a research proposal that contains the followings:

1. Determine **TWO (2)** objectives for your proposal.

(4 marks)

2. Identify **TWO (2)** possible theories that can be adopted for this research.

(2 marks)

3. Assuming you are doing a **quantitative study** for this research:
   a) Describe your sample and sampling approach.

(4 marks)

   b) Describe your data collection process.

(2 marks)

   c) Suggest **TWO (2)** types of analysis you can perform for the proposed study.

(4 marks)

4. Alternatively, if you opt to use qualitative approach, propose **TWO (2)** interview questions to address the research objectives stated in Question 1.

(4 marks)

CASE 3

In a suburban community, a market survey was conducted to introduce SmartHealth, a new health monitoring device capable of tracking vital signs and communicating health data to smartphones or healthcare systems. Out of 600 identified households, questionnaires were distributed, yielding a 25% response rate, and these responses were obtained as a valid sample. A normality test was employed to assess whether the data set was well-modelled by a normal distribution. The test on resident age indicated a normal distribution. Data analysis revealed that the majority positively intended to purchase the device. Then, correlation analyses were performed to explore relationships between variables. The analyses on resident age and income unveiled weak positive and moderate positive correlations with the intention to purchase.

Additionally, a cross-tabulation analysis was conducted to analyse categorical (nominal measurement scale) data. The cross-tabulation revealed that higher-income young professionals exhibited the highest intent, while older residents, particularly retirees, expressed lower interest. These findings suggest that marketing efforts should target both demographics, emphasising different benefits and pricing strategies based on income levels. Education campaigns may be crucial for addressing concerns and increasing adoption among older residents.

Answer the following questions based on the case studies:

1. Identify the unit of analysis.

(2 marks)

2. Indicate the size of the sample.

(2 marks)

3. Define the population.

(2 marks)

4. As revealed by the cross-tabulation results, provide **TWO (2)** elaboration on how marketing strategies can effectively cater to the diverse preferences and intentions observed in different demographic groups, such as young professionals with higher incomes and older retirees.

(4 marks)

5. Evaluate the suitability of snowball sampling as a method for gathering data in the suburban community case study.

(2 marks)

6. Explain the most suitable sampling method for the suburban community case study.

(2 marks)

7. Based on the answer given in Question 6, provide **THREE (3)** justifications.

(6 marks)

CASE 4

There are five data collection methods. Identify **ONE (1)** method used in each case and provide **TWO (2)** explanations for each identified method.

Case	Data collection method	Explanation
A researcher quietly watches students in a classroom to see how they work together during group activities. The researcher notes what the students do and how they talk to each other. This helps the researcher understand how students interact and learn in groups.		
A researcher talks to employees individually to understand their satisfaction with their jobs. The sessions involve open-ended questions, letting employees share their feelings and experiences. This helps the researcher learn more about what factors contribute to job satisfaction.		

Researchers distribute a set of questions to a large group of people to gather insights into their preferences for a new product. The goal is to collect diverse opinions and understand what features are most valued by users.		
Researchers systematically analyse and categorise the themes and patterns present in a collection of online reviews for a new product. By scrutinising the language used and identifying recurring topics, they aim to gain insights into consumer sentiments, preferences, and areas of improvement associated with the product.		

(20 marks)

CASE 5

A disruptive innovation in this field applies to cryptocurrencies. Advocates of these new means of payment claim various advantages, like a fully decentralised, peer-to-peer transaction system, elimination of chargeback risks, lower associated transaction costs, increased level of security, greater ease of use, and full support for mobile devices. Of the several hundred different cryptocurrencies, Bitcoin is the most well-known, most discussed.

To the best of our knowledge, there has not been any empirical research about factors influencing bitcoin adoption. With this paper, we contribute to this under-researched turf by means of an exploratory, qualitative approach using 13 semi-structured interviews and consequently using quantitative approach for validating the model developed.

Below are some excerpts of Interview transcripts regarding factors influencing Bitcoin adoption. To analyse the interview scripts, several codes have been formulated. Below are the descriptions of the codes:

a) **Perceived ease of use** is defined as the degree to which an individual believes that using a particular technology would be free from effort.
b) **Perceived risk** is defined as the perceived uncertainty in a situation.
c) **Perceived usefulness** is defined as people's perceptions of whether a new technology could change the way they do business, and whether it could make their performance better.

Following the interview analysis, factors influencing bitcoin adoption were identified and survey was distributed to examine the relationship between the factors and the intention to adopt bitcoin.

Survey invitation has been distributed to several Bitcoin community groups. In total, 299 respondents answered the survey. Analysis result demonstrates perceived risks significantly influence perceived ease of use, while perceived ease of use significantly influences perceived usefulness. In addition, perceived usefulness significantly related to Intention to use.

1. Based on the codes' definition above, please categorise the following interview excerpts to the relevant codes:

Interview Transcripts	Code
B3: "the volatility of Bitcoin and the frictions of moving money in and out of Bitcoin are a limiting factor in its adoption.	

C2: "It was really complicated and not easy to use"	
B2: "a secure and efficient, trustful decentralised monetary transfer system, perfect for investment"	
B1: "for merchants, Bitcoin displays a direct money inflow of their revenues, whereas with CC payment they need to wait for around a week to receive their payments	
B1: "paying with Bitcoin is both convenient and secure. We just need to scan a QR code in these days and our payments are transferred successfully. Anywhere, anytime"	
B3: "System like Bitcoin is needed to increase speed and decrease fees at the same time"	

(12 marks)

2. Based on the factors identified above, draw the appropriate research model of the study.

(5 marks)

3. Provide **THREE (3)** examples of descriptive statistics for the study above.

(3 marks)

4. Write an abstract to explain about the Bitcoin study. The abstract should contain issues, objectives, methodology, findings, and contributions, and not more than **150 words**.

ABSTRACT	
Issues	
Objectives	
Methodology	

Findings	
Contributions	

(10 marks)

END OF QUESTIONS